

Course Outcomes (CO):

On completion of the course, the student will be able to

CO1: Understand the principles of mechanics (statics and dynamics), the concept of free body diagrams, and resolution of forces.

CO2: Apply the principles of mechanics, concept of free body diagrams, resolution of forces, and equations of equilibrium or motion to a given engineering or physical application.

CO3: Analyse a given engineering problem or physical applications and calculate the required parameters like forces, moments, displacement, velocity, acceleration, etc.

CO4: Appreciate the physical and mathematical meaning of quantities like centroid and moment of inertia and their applications in engineering and locate the centroid and calculate the moment of inertia of different sections used in different engineering applications.

Bloom's Taxonomy Levels (BL)

Remember	Understand	Apply	Analyse	Design	Create
L1	L2	L3	L4	L5	L6

Programme Outcomes (PO)

PO1-Engineering Knowledge	PO2-Problem Analysis	PO3-Development of solutions	PO4- Conduct investigation of complex problems
PO5- Modern tool usage	PO6- Engineer& Society	PO7-Environment & Sustainability	PO8-Ethics
PO9-Individual& team work	PO10-Communication	PO11-Project management &finance	PO12-Lifelong learning

Programme Educational Objectives (PEO)

PEO1	Preparation: To prepare the graduates or to succeed in industry/technical profession through global rigorous education.
PEO2	Core Competence: To provide the graduates with a solid foundation in mathematical, scientific, and engineering fundamentals required to solve engineering problems and also to pursue higher studies
PEO3	Breadth: To train the graduates with good scientific and engineering breadth so as to comprehend, analyse, design, and create novel solutions and products for the real-life problems
PEO4	Professionalism: To inculcate in the graduates, a professional and ethical attitude, effective communication and team work skills, a multi-disciplinary approach, and an ability to relate engineering issues to broader social context
PEO5	Learning environment: To provide the graduates with an academic environment making them aware of excellence, leadership and teamwork skills, ethical codes and guidelines, and life-long learning needed for a successful professional career

Programme Specific Outcomes (PSO)

PSO1	The graduate should excel in product design, thermal and fluid engineering, and manufacturing systems with the acquired knowledge in mathematics, science, and engineering disciplines.
PSO2	The graduate should be competent to analyze, interpret, and solve real life mechanical engineering problems
PSO3	The graduate should have the managerial skills to work effectively in a team and in a society by following ethical and environmental practices

Mapping of Course Outcomes with Programme Outcomes and Programme Specific Outcomes

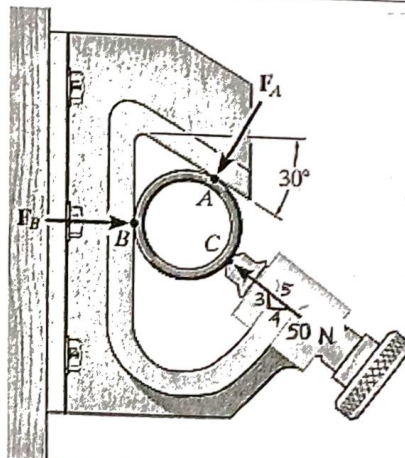
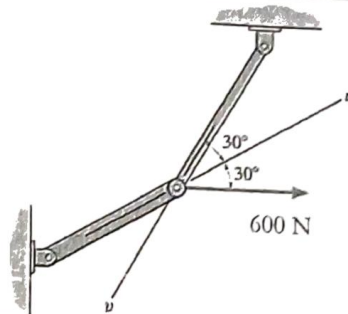
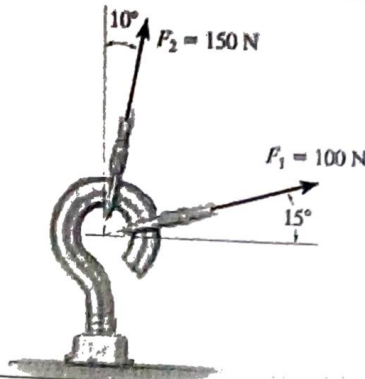
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	1				1				1	3	3	
CO2	3	3	1	3				1				1	3	3	
CO3	3	3	1	3				1				1	3	3	
CO4	3	3	1	1				1				1	3	3	

Mapping of Questions with Course Outcomes**Mapping of Questions with Blooms Taxonomy levels**

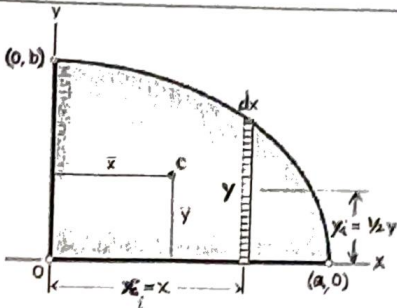
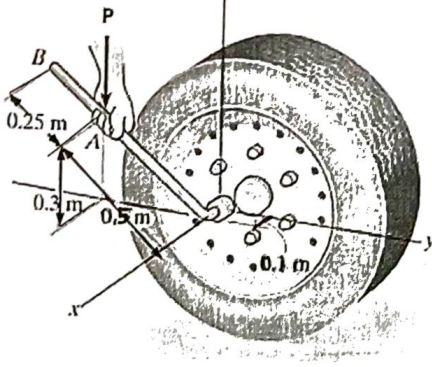
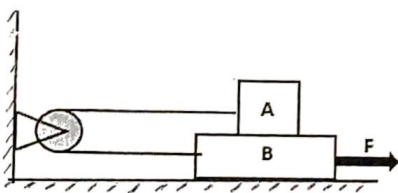
	CO1	CO2	CO3	CO4		L1	L2	L3	L4	L5	L6	
Q1	10	7	3			5	10	5				
Q2				10					10			
Q3		5	5					10				
Q4	4	3	3						10			
Total	14	15	11	10		5	10	15	20			
%	28%	30%	22%	20%		10%	20%	30%	40%			

PART A

No	Question	CO	BL	Marks
1. a	State and prove the Varignon's theorem of moments	CO1	L1	5
b	The screw eye in the figure is subjected to two forces F_1 and F_2 . Determine the magnitude and direction of the resultant force	CO1	L2	5
c	Resolve the horizontal force 600 N in to components acting along the u and v axes and determine these components	CO2	L2	5
d	The pipe is held in place by the vice. The bolt exerts a force of 50 N on the pipe in the direction shown. i) Resolve the forces in x-y directions. ii) Determine the forces F_A and F_B that the smooth contacts at A and B exert on the pipe	CO2 CO3	L3	2 3



PART B

No	Question	CO	BL	Marks
2	Locate the centroid of the quarter elliptical area bounded by the x-y axes and the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 	CO4	L4	10
3	A vertical force $P = 100 \text{ N}$ is applied at A on the spanner to remove the nut on the wheel of the automobile. i) Determine if this force is adequate, provided a torque of 50 N.m about the x-axis is initially required to turn the nut. ii) If the force 100 N can be applied at A in any other direction, will it be possible to turn the nut? 	CO2	L3	5
4	The system of two blocks, cable, and fixed smooth pulley is initially at rest. The coefficient of friction is 0.5 between the blocks and 0.6 between the block and floor. The weight of the block A is 5 kN and that of block B is 10 kN. i) Draw the free body diagrams of blocks A and B. ii) Write down the equations of equilibrium iii) Determine the horizontal force F necessary to cause motion and the tension in the cable. 	CO1	L4	4
		CO2		3
		CO3		3